

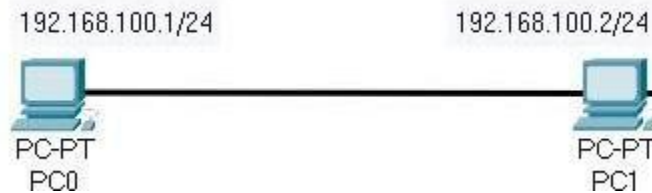
CO323 - 2025 (E/21)

Continues Lab Sheet

General Instructions:

1. For all the communications that you initiate between devices, ensure that you observe the entire packet transfer through a packet-capturing device, such as **Wireshark**.
2. As you move through the tasks, it may be necessary to switch between **promiscuous** and **non-promiscuous** modes to distinguish between *what you see* and *what you opt to read*. [we'll try to hint to you when necessary, but you are free - & encouraged to - ask the question yourself & test those out if in doubt]
3. Testing as it is VS. clearing caches (e.g., clearing ARP cache in devices, resetting the switch, restarting the switch (or clearing data)) will have different effects on the results. Try all possibilities at the beginning. Later, you'll know what to do & when.

Task1: Assume a situation where you do not have access to the internet and connect your PCs through an RJ45 direct cable connection to share some files.



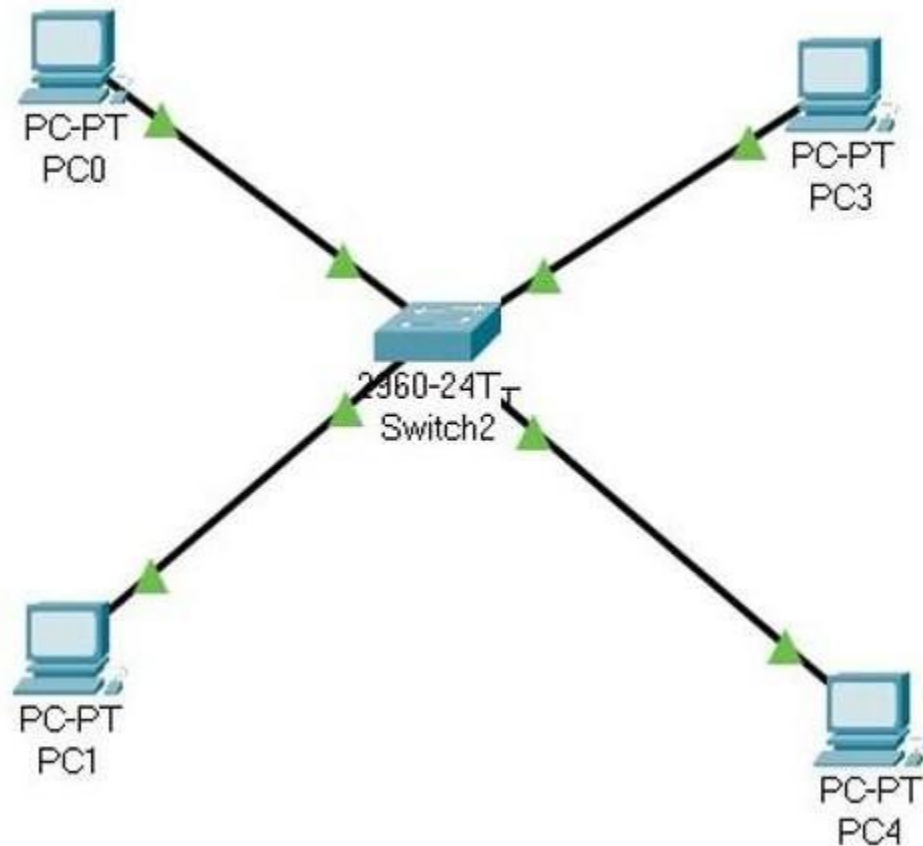
1. **Connect two PCs through a RJ45 direct cable as shown above.**
 - a. Assign IP addresses and subnet masks.
 - b. Open a command prompt at both of the PCs and try to ping the other. What do you observe? Explain your observations.
 - c. Change the cable to a crossover cable and check the connectivity
 - d. Discuss the observations among the group members.

Check the following combinations of IP addresses and Subnet masks and explain the results.

PC0 IP address	PC0 Subnet mask
192.168.200.1	24
192.168.200.1	16
192.168.100.10	28
192.168.100.250	28

2. **Connect the 2 PCs through a Switch and do the same experiment.**

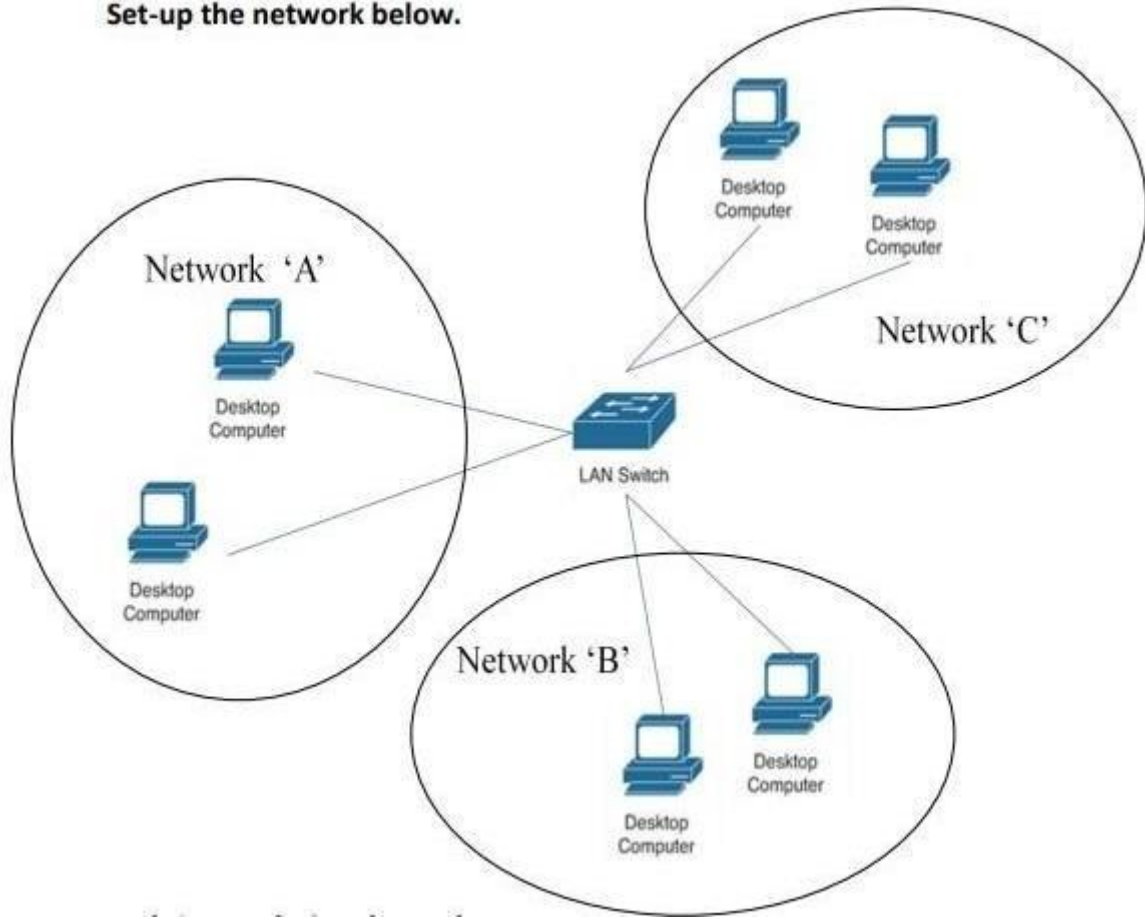
Task 2: Connect Multiple PCs through a Switch, in the different configurations and scenarios given below and observe the results (both direct communications and Wireshark captures)



- 1) Assign all PCs to the same subnet (allocate appropriate IP addresses & Subnet masks)
 - a) Ping between all devices
 - b) Change the IP address of One of the PCs (say C) and observe the effect
 - First ping from other devices to C - what happens?
 - Then ping from C to others - what happens? Is there any difference?
 - Now ping again from other devices to C - is there any difference?
 - c) Change the switch port of one PC (say A)
 - First ping from other devices to A - what happens?
 - Then ping from A to others - what happens?
 - Now ping again from other devices to A - what happens?

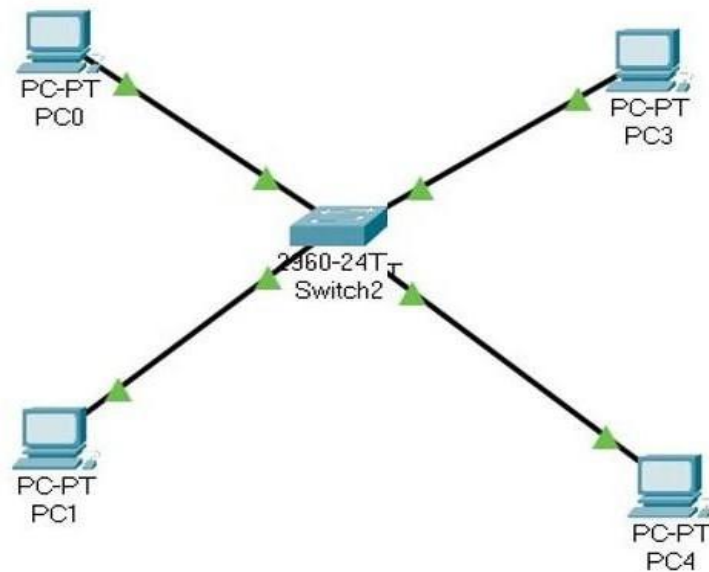
- 2) Assign the PCs to different subnets, with at least one pair of PCs in a given subnet.

Set-up the network below.



- Ping between the devices in each subnet (observe the Wireshark captures in both modes - hint: Do this after restarting the switch & while in operation)
- Change the subnet of different PCs (e.g., configure one PC from network B to belong to network A, and observe the effect).

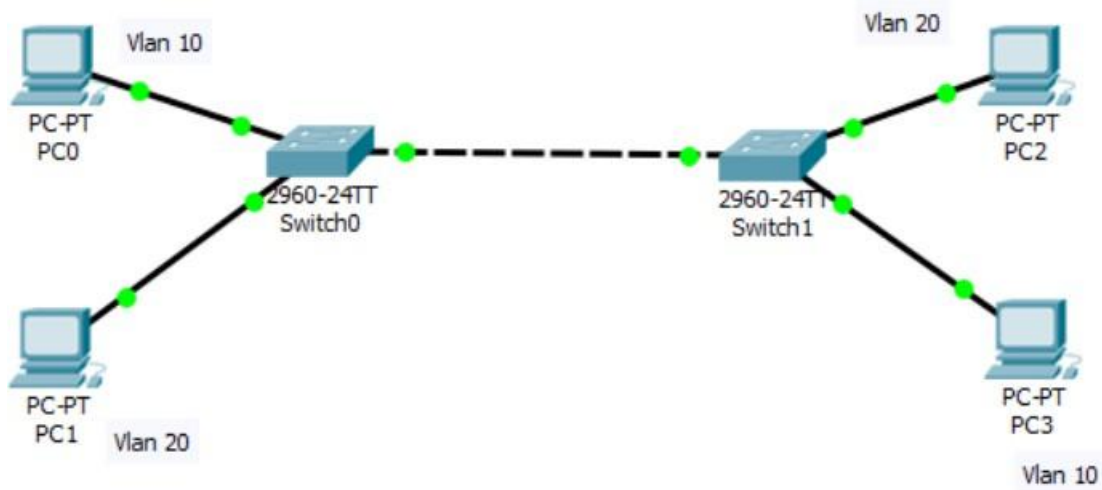
Task 3: Connect Multiple PCs through a Switch, in the different configurations and scenarios given below



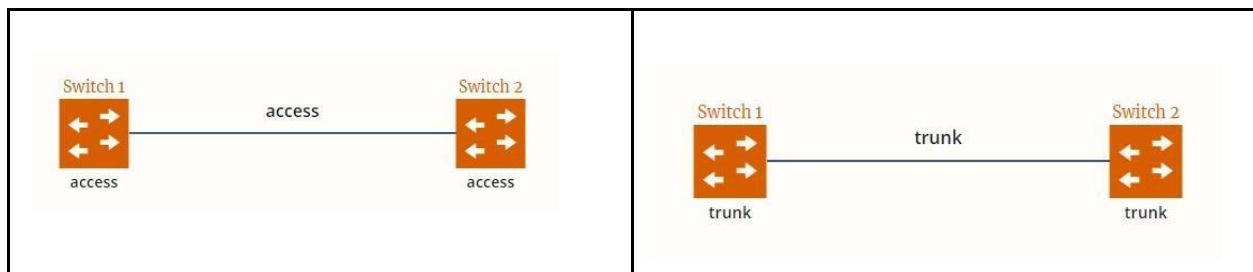
Assign all PCs to the same subnet (allocate appropriate IP addresses & Subnet masks) as in task 2.

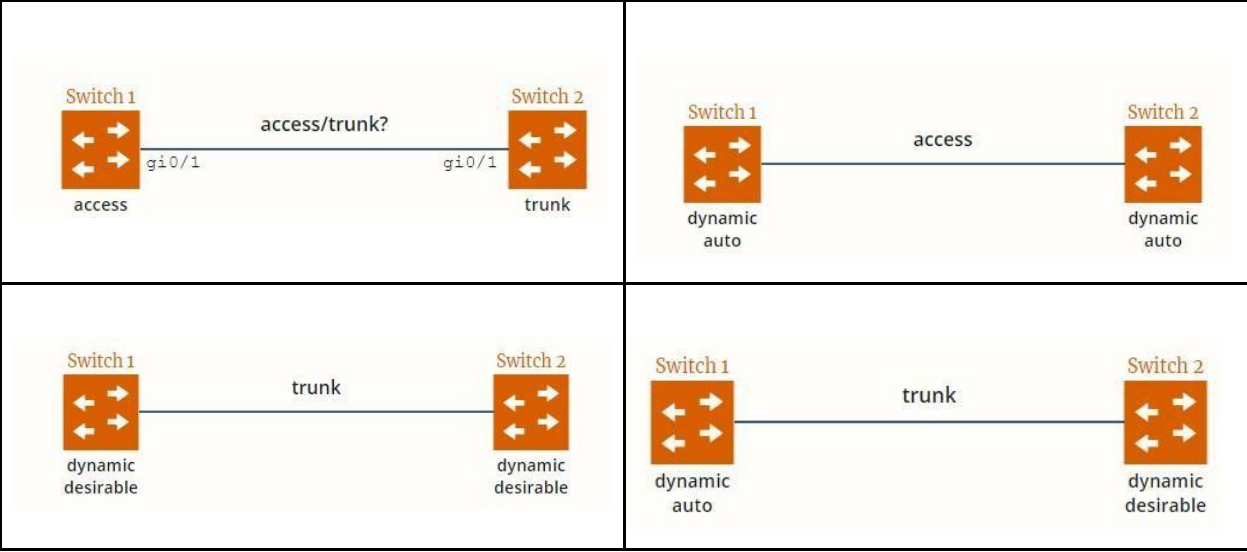
1. Define 2 different VLANs (PC0 and PC1 in VLAN10, PC2 and PC3 in VLAN 20).
2. Observe the differences between the 2 settings (The same network with and without VLAN) using Wireshark.
 - a. Discuss whether PCs in different VLANs should be assigned IP addresses in the same subnet.
3. Allocate PCs in separate VLANs with IPs in different subnets (e.g., PCs in VLAN10 given 192.168.10.0/24 and PCs in VLAN20 given 192.168.20.0/24), and observe the performances.
 - a. Discuss the implications of assigning different subnets to different VLANs. Is this necessary?
4. Connect two ports belonging to the two VLANs with a cable and observe the packet transfer.
 - a. Discuss the security concerns of the above method.
5. Allocate multiple (4 or more) PCs to a single VLAN. Allocate those devices with IP addresses in different subnets, with at least two PCs in a given subnet.
 - a. Observe the communication between devices.
 - b. Should devices in a given VLAN be allowed to assign any IP address they want?

Task 4:



1. Configure the following network, but all in the same VLAN (e.g., VLAN10)
 - a. Configure the ports connected with dotted lines as access ports.
2. Assign IP addresses and subnet masks according to the following scenarios and observe the communication (i.e., who can communicate, and who will see others).
 - a. All in Single subnet
 - b. PC0/PC2 and PC1/PC3 in 2 different Subnets, respectively
3. Configure the VLANs according to the figure above, and the dotted line as a “trunk port” with the following details. Record your observations.
 - a. VLAN10 untagged, and VLAN20 tagged
 - b. VLAN10 tagged, and VLAN20 untagged
 - c. Both VLAN10 and VLAN20 tagged
 - d. Both VLAN10 and VLAN20 untagged
4. Research and learn the difference between an access port and a trunk port.
5. Observe the traffic in the following port modes



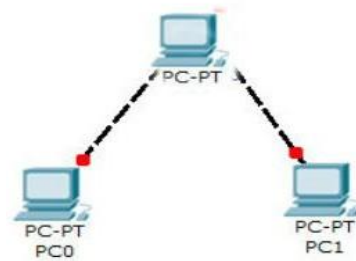


Task 5: Basic Routing

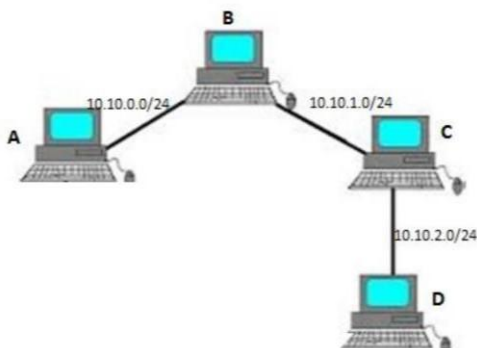
Part A

In this network, 3 PCs are used and the 2nd(middle) PC works as a router.

1. Using the above setup configures the static routing.

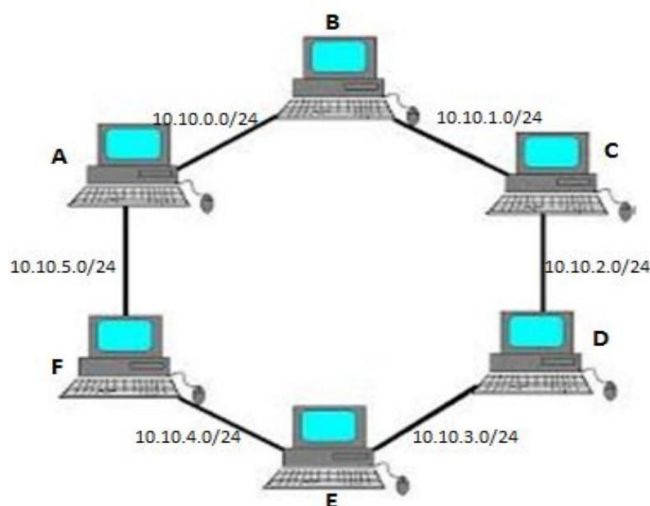


Part B



1. Use the dual ethernet-enabled PC and implement the network setup shown above by configuring the PCs with static routing.
2. Try to communicate from A to D. Observe the Routing tables at B and C

Part C



1. Use the dual ethernet-enabled PCs and implement the network setup shown above and configure the PCs using static routing.
2. Set the configurations to communicate from A to D in the clockwise direction. Identify the request/response path of the communication and the route.
3. Set the configurations to communicate from A to D in the anti-clockwise direction. Identify the request/response path of the communication and the route.

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